

NIRANJ SANTHOSH

UNDERGRADUATE PHYSICS STUDENT

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OBJECTIVE

To build a solid background in physics through a research internship in astrophysics or related fields, focusing on contributing to major projects and scientific studies.

WORK & LEADERSHIP EXPERIENCE

Sky Secrets | Astronomy Head & Science Communicator | January 2025 - Present

- Delivered astronomy focused lectures with hands on telescope demonstration in schools and colleges across Chennai, advocating for evidence based reasoning and scientific method.
- Conducted community stargazing events in urban and rural areas, bridging the accessibility gap in space science.
- Designed and calibrated a custom RTL-SDR radio telescope and conducted testing of the antenna gain and receiver sensitivity.

India Space Academy | Project Intern | June 2025 - July 2025

- Finished a 120-hour intensive program focused on data-driven astronomy and computational research.
- Calculated the dynamical mass of a galaxy cluster using SDSS spectroscopic data to identify dark matter discrepancies and calculate their velocity dispersion.

International Astronomical Search Collaboration | Participant | April - Aug 2025

- Used Astrometrica to analyze Pan-STARRS data for detecting and tracking Near-Earth Asteroids, resulting in 3 team-based provisional detections.

Loyola Astronomy and Innovative Forum | Telescope Operator | June 2025 - Present

- Managed the setup, calibration, and routine maintenance of departmental telescopes, while leading the observation sessions.

Indian Centre for Space Physics | Participant | December 2025

- Gained hands on experience in optical astronomy data analysis (IRAF), black hole model fitting using TCAF and radio signal propagation simulations.
 - Conducted Monte Carlo Simulations using GEANT4 to model instrument performance for balloon borne astronomy.
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EDUCATION

B.Sc Physics | 2024 - Present

Loyola College, Chennai, Tamil Nadu

CGPA : 9.5

Relevant Coursework

- Classical Mechanics
- Astronomy and Astrophysics
- Electricity and Magnetism
- Mathematical Methods for Physics
- Properties of Matter
- Electronics
- Statistical Methods in Physics (Python)
- Chemistry for Physics

SKILLS

- Technical skills :** Python · MATLAB · Mathematica · SQL · DS9 · Astrometrica · LaTeX · GQRX · PyRAF · DAOPHOT · Telescope Operation
- Soft Skills :** Scientific Communication · Problem-Solving · Technical Writing · Leadership · Teaching · Analytical Thinking.
- Languages :** English (Fluent) · Hindi (Fluent) · Malayalam & Tamil (Native) · German (Elementary).